

SANYA LIUDAOLING, China

WMO: 599480

Lat: 18.2256N

Lon: 109.5914E

Elev: 1378

StdP: 13.98

Time Zone: 8.00 (E08)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 54.0	(c) 56.5	(d) 44.4	(e) 45.3	(f) 55.8	(g) 48.2	(h) 52.6	(i) 58.5	(j) 32.3	(k) 62.5	(l) 30.3	(m) 62.9	(n) 16.8	(o) 40	(p) 0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 6.5	(c) 85.9	(d) 79.4	(e) 84.8	(f) 78.9	(g) 83.9	(h) 78.5	(i) 80.7	(j) 83.9	(k) 80.0	(l) 82.9	(m) 79.4	(n) 82.1	(o) 7.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.8	163.0	82.2	79.2	159.4	81.4	78.7	157.1	81.0	45.7	84.2	44.9	83.4	44.3	82.5	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 29.2	(b) 25.8	(c) 23.3	(e) 50.2	(f) 88.0	(g) 3.1	(h) 1.2	(i) 48.0	(j) 88.9	(k) 46.1	(l) 89.6	(m) 44.4	(n) 90.3	(o) 42.2	(p) 91.2		
(a) 29.2	(b) 25.8	(c) 23.3	(e) 44.9	(f) 81.4	(g) 2.4	(h) 1.1	(i) 43.2	(j) 82.2	(k) 41.8	(l) 82.8	(m) 40.5	(n) 83.4	(o) 38.7	(p) 84.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.1	65.8	67.8	71.0	75.4	78.9	80.0	78.8	78.7	77.9	74.9	72.2	67.4		
	DBStd	5.71	3.98	4.28	3.70	2.76	2.24	1.97	2.37	2.02	2.04	2.09	2.67	4.15		
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	96	37	22	7	0	0	0	0	0	0	0	2	28		
	CDD50	8792	491	498	650	761	897	899	894	888	838	772	665	539		
	CDD65	3413	63	100	192	311	432	449	429	423	388	307	217	102		
	CDH74	18714	13	55	257	1214	3163	3881	3247	3105	2457	980	297	45		
	CDH80	2319	0	0	2	71	482	638	460	360	268	37	1	0		
(14) Wind	WSAvg	11.7	15.1	12.4	11.8	10.4	8.4	8.6	8.8	9.3	10.2	13.7	16.1	15.8		
(15) Precipitation	PrecAvg	62.7	0.4	0.4	0.8	2.6	5.2	7.2	10.1	9.4	11.9	10.5	3.2	0.9		
	PrecMax	87.1	2.2	1.9	4.5	6.7	14.2	16.8	24.2	24.5	25.3	33.8	10.8	5.6		
	PrecMin	40.0	0.0	0.0	0.0	0.0	0.6	2.0	4.3	1.9	4.7	1.3	0.0	0.0		
	PrecStd	12.1	0.5	0.5	1.1	1.9	3.4	3.3	4.3	4.7	5.7	7.2	3.3	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.9	77.5	79.8	83.8	87.8	87.0	87.0	85.6	84.8	82.7	79.7	77.0		
		MCWB	70.7	71.5	73.5	76.4	79.3	79.5	79.6	79.8	78.8	75.7	72.6	71.6		
	2%	DB	73.3	75.6	78.1	81.8	85.5	85.9	85.2	84.3	83.6	81.0	78.2	75.3		
		MCWB	68.3	70.9	73.0	76.0	78.9	79.5	79.0	79.0	78.3	74.5	72.9	70.6		
	5%	DB	71.9	74.2	76.7	80.6	84.1	84.6	83.9	83.2	82.7	79.8	77.0	73.7		
		MCWB	67.9	70.2	72.5	75.8	78.4	79.0	78.4	78.4	77.7	73.8	72.5	69.6		
	10%	DB	70.6	72.8	75.3	79.2	82.7	83.6	82.6	82.1	81.7	78.6	75.8	72.4		
		MCWB	67.3	69.6	72.0	75.3	77.9	78.6	78.0	77.8	77.2	73.3	71.9	68.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	72.9	75.4	78.1	81.4	81.3	80.9	81.4	80.7	77.7	75.4	72.8		
		MCDB	74.2	75.5	77.4	81.0	84.9	84.7	84.7	83.9	83.2	80.8	77.0	75.0		
	2%	WB	70.5	72.0	74.3	77.3	80.1	80.5	79.9	80.1	79.6	76.4	74.5	71.8		
		MCDB	71.7	74.0	76.2	79.8	83.1	83.8	83.3	82.7	82.1	79.0	76.2	73.7		
	5%	WB	69.2	71.1	73.5	76.5	79.4	79.9	79.2	79.2	78.8	75.5	73.8	70.9		
		MCDB	70.7	72.9	75.3	78.8	82.0	82.9	82.1	81.5	81.3	77.7	75.4	72.7		
	10%	WB	68.2	70.4	72.7	76.0	78.7	79.3	78.6	78.7	78.1	74.7	73.1	70.0		
		MCDB	70.0	72.2	74.4	78.1	81.1	81.9	81.3	80.8	80.4	76.8	74.8	71.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.8	7.2	7.2	7.5	6.5	6.4	6.3	6.9	7.0	6.7	7.7		
		MCDBR	7.9	8.0	7.9	8.0	8.8	7.9	7.9	7.6	8.3	8.5	7.7	7.8		
		MCWBR	4.4	3.7	3.6	3.5	4.1	4.2	4.3	4.7	4.7	4.8	4.4	4.8		
	5% WB	MCDBR	6.3	6.9	6.4	7.1	7.8	7.3	7.2	6.9	7.3	6.9	6.1	7.1		
		MCWBR	4.1	3.5	3.3	3.7	4.2	4.3	4.3	4.6	4.7	4.3	3.7	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.467	0.481	0.571	0.541	0.450	0.440	0.423	0.444	0.492	0.525	0.477	0.492		
	taud		2.111	2.110	1.895	2.037	2.360	2.406	2.449	2.373	2.203	2.045	2.158	2.046		
	Ebn at Noon		257	261	241	249	270	270	275	272	259	245	252	244		
	Edh at Noon		48	50	64	56	40	38	36	39	46	53	45	49		
(44) All-Sky Solar Radiation	RadAvg		1166	1370	1495	1748	1847	1735	1705	1626	1523	1422	1269	1093		
	RadStd		146	190	113	136	121	143	164	115	157	171	109	117		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.47	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	+162	+149

Nomenclature: See separate page